WormBot: Performing In-pipe Locomotion using soft Origami Pneumatic Actuators

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ROB 498 Final Project Report

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Abstract:

This paper explores the development of a soft-pneumatic actuator for the purpose of competing in the Robosoft In-pipe Locomotion competition. Traditional rigid robots can often struggle in tight, cramped environments opening up the need for more research of soft robotic solutions for these applications. Our proposed design utilizes three connected kresling origami actuators which can be individually controlled using pneumatics. The final design proved to be fairly successful when tested in the Robosoft competition, able to traverse through all sections but the rubble and sand pit obstacles. There is still a need for adjustment to allow for more flexibility to deal with obstacles when traversing and to ensure reliable operation especially through thorough sealing of all pneumatic connections. While adjustments do still need to be made, this project highlights the potential for soft robotic solutions such as this design in pipeline traversal applications.

1. Introduction

The field of soft robotics has demonstrated great potential with applications that require adaptability with delicate environments. Traditionally rigid robots struggle with moving through confined spaces and flexible pipelines, particularly when the paths are cluttered with obstacles or deformations in the structure [1]. Many designs rely on wheels which may not function properly on various terrains such as sand and rubble [2]. The purpose of this work is to develop and improve alternative methods for soft robots to navigate through cluttered pipelines through the usage of the Robosoft competition.

Pipe-crawling robots, such as ones demonstrated in this competition, have been researched widely for their many applications for pipeline systems, including inspection [3], maintenance [4], and cleaning tasks [5, 6]. These spaces are typically inaccessible by humans and require small robots capable of traversing pipeline systems. Many researchers have attempted to solve this problem. Liu developed a tensegrity robot capable of horizontal and vertical locomotion in pipelines [7]. Wang created a soft robot using pneumatic actuators made of braided origami silicon to climb through the pipe. The origami has a unique design and each section can be actuated individually to facilitate turning [8]. Additionally, Ze et al. used magnetic actuators to drive a soft robot through a pipe [9].

Our proposed method consists of an entirely soft structure and three pneumatic actuators (Fig 1.), connected to end caps containing feet, which drive the robot forward using friction. Because the robot's structure is fully soft, it is able to move through various surfaces and obstacles that are unplanned or unexpected, such as dents and rubble.

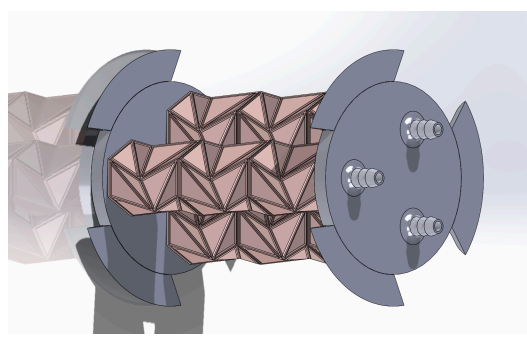


Fig. 1: Image of the proposed design solution in CAD modeling software.

This project is intended to complete the Robosoft In-pipe Locomotion competition with as high of a score as possible. Each section of the pipe gives different points depending on how many tries it takes to complete, and the perceived difficulty of the obstacle. The layout of the pipe can be seen in Figure 2.

In the robosoft competition, we expect to be able to drive our robot through all straight segments, and all ninety degree turn segments. We are unsure if the design is capable of navigating the rubble, flexible wall, and sandpit segments. In the case of failure on these segments, further design iterations could modify the friction coefficients, rigidity, size and other parameters to facilitate further flexibility. A successful solution should be able to move forward and turn in the pipeline consistently. The competition places a 10 minute time limit on each competing robot and a size constraint of having to fit within the tube's 5 cm internal diameter..

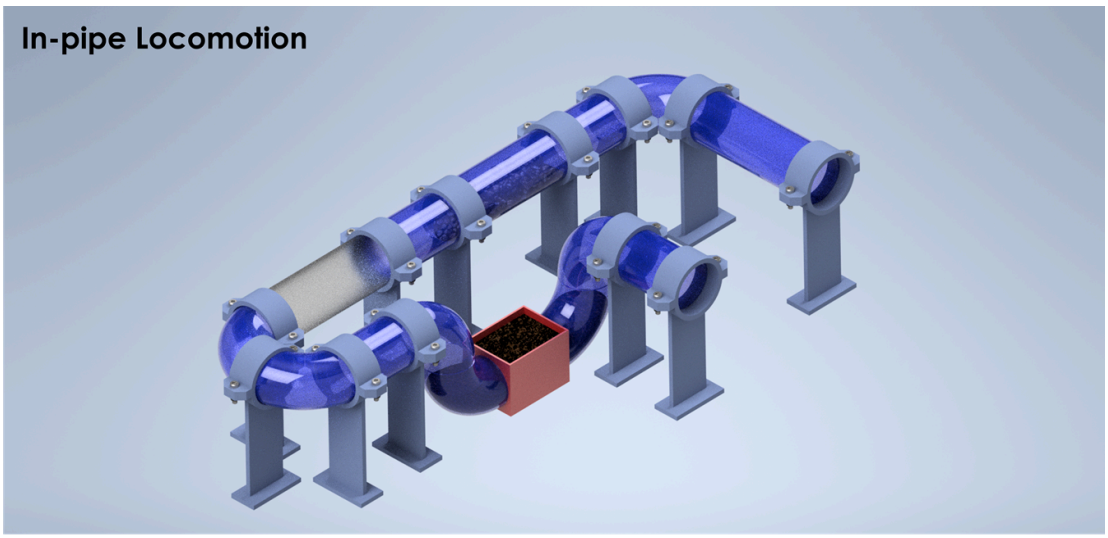


Fig. 2: Schematic of the Robosoft In-pipe Locomotion challenge layout. Taken from robosoft2025.org

2. Methodology

2.1 Design Overview

Our initial design solution was the use of a pneumatic actuated system with a kresling origami structure to ensure linear motion. The idea to use kresling origami was inspired by the work of Ze et al. who created a similar robot using a kresling structure actuated through magnetic plates as seen in figure 3. Their project saw the successful creation of a "crawling" robot using this design and were able to create a forward linear motion when actuated. This design used a magnetic field around the robot to allow for steering capabilities, but this feature would not be feasible in the implementation of our design. We also opted for the use of pneumatics as the method of actuation rather than magnetism due to the ease of control and because our team had little to no experience working with pneumatics and wanted to learn more about them.

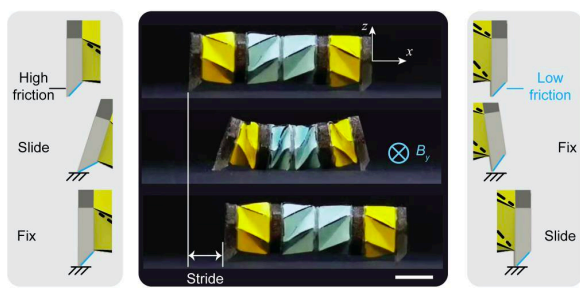


Fig. 3: Kresling origami crawler created by Ze et al.

Another feature we intended to add to our initial design was the addition of kirigami "snakeskin" around the actuator to facilitate forward motion. This feature was inspired by the work of Rafsanjani et al. which created a pneumatic actuator using this kirigami "snakeskin" as seen in figure 4. This snakeskin allowed for different coefficients of friction between forward and backward motion allowing for a linear forward motion. Rafsanjani et al. tested this design with different kirigami patterns and found that the most effective was a trapezoidal pattern. In the context of our project, we were hoping

that the addition of this feature would allow our robot to have reliable linear forward motion while progressing through the course while not inhibiting its ability to traverse through obstacles. Our original plan was to create this snakeskin by cutting the kirigami pattern into a thin sheet of plexiglass plastic using a laser cutter.

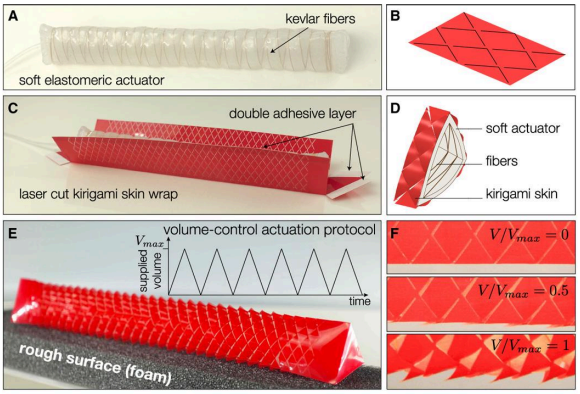
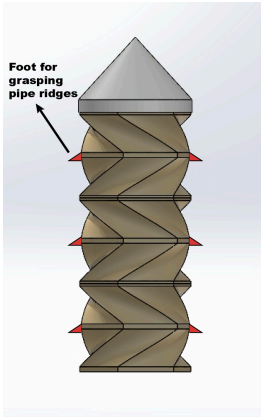


Fig. 4. Pneumatic actuator with kirigami snakeskin design.

After seeing the tubes being used for the course we decided to make some modifications to our design. This decision came primarily from the realization that the tubes contain ridges along them which may allow for an easier method of facilitating forward motion. To compensate for this, we decided to change our design to use triangle-shaped feet along the sides of the robot rather than kirigami snake skin in hopes that the feet would latch on to the ridges along the tubes preventing backward motion while still allowing for forward motion. After manufacturing, the end design is the prototype shown in figure 5. This prototype was completely unsuccessful with absolutely no contraction when connected to a vacuum pump. We were able to track down this issue to three primary causes: a lack of certain folds in the origami design, too thick of an origami structure leading to stiffness in the print, and a faulty vacuum pump.

In our second prototype we made these changes and began to see some contraction in our prototype but still not a sufficient amount meaning that we needed to use an even less stiff material and further reduce the thickness of the walls of the part. Before making this prototype we also made some changes to the



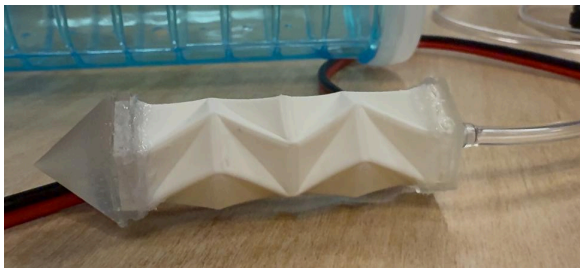


Fig. 5. *Top:* CAD model *Bottom:* Fabricated model of initial design without feet

direction than the other. We also removed the cone from the front cap as we learned that we would no longer need to burrow for the sandpit section of the course. After making these changes to the end caps and stiffness we had a prototype, figure 6, that had sufficient contraction, but found that the design of the feet was too stiff to allow bending past the ridges.

Fig. 6. Third prototype with successful contraction and addition of feet.

In our next prototype we slightly adjusted the feet to make it easier for them to pass the ridges by adding a chamfer along the front side, reducing the thickness, and removing the teeth. We found that this prototype

finally had success in traversing the course. We were able to see reliable forward linear motion through the straightaway sections of the course but realized that the robot struggled to move at all once reaching the turn sections. Our solution to this problem was to shrink the size of the actuator and put three of these actuators together in a triangle formation which can each be actuated individually. After making the necessary adjustments to the physical design as well as the code and circuit used to control the actuators we ended up with our final design shown in figure 1 and figure 7.

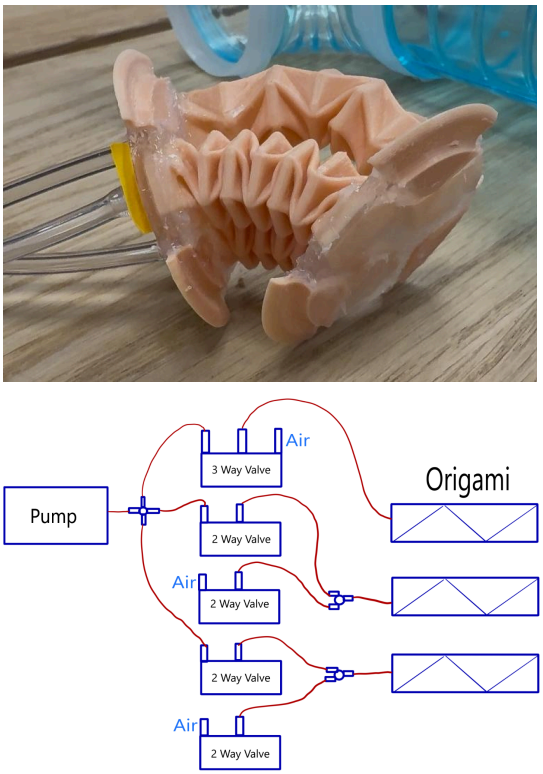


Fig. 7: *Top:* Final prototype using three individually controlled pneumatic actuators. *Bottom:* Electro-pneumatic circuit diagram

The actuator was controlled using an electro-pneumatic circuit as shown in figure 7. A vacuum pump was connected to a pneumatic cross connector splitting the flow into three outputs, one for each actuator. Ideally the circuit should be using a three-way valve solenoid to control each origami, but due to a lack of resources we had to instead use two two-way valves to control two of the actuators. While this made the physical circuit and code used more complicated, it did work in exactly the same way. These valves worked where when we sent a signal to a valve from a digital pin of an arduino, it would go through a relay to regulate the current and voltage, then into the valve triggering the actuator to either be connected to the vacuum for contraction, or connected to the atmosphere to release. This system was controlled with arduino code which allowed each actuator to individually contract when specific inputs were sent to the arduino. The software also allowed for a global actuation command input which would contract all the actuators at the same time. Pairs of actuators could be actuated simultaneously by inputting their specified commands together. This system allowed movements in 6 different directions, as well as the inchworm-like movement by actuating all three origami simultaneously.

2.2 Fabrication and Experimentation Methods

In the fabrication process of our designs, a large majority of the parts being used were fabricated using different 3D printing materials. For the soft parts of the robot we used TPU and tested with a variety of different stiffnesses finding TPU 70A to be the best for our application, printed using a Bambu FDM printer. For the rigid end cap parts we used a clear resin material printed with a Formlab 4 SLA printer. The use of 3D printed parts allowed for a quick turnaround between prototypes as we were able to quickly make updates to our design and print new versions rather than having to wait for parts to be manufactured professionally or bought and shipped between prototypes.

Our experimentation consisted of a basic testing setup just using a small portion of the full course with just one straight section and one 90 degree turn section. We were able to adjust the orientation of this setup to also allow for basic testing of the sections where we need to climb down and up within the course. We also did some basic testing simply placing the robot on a table, but were only able to test contraction and controls there as our design required the ridges of the tubing for forward motion. In our experimentation we were unable to get to the point of characterizing the actuation and motion of our robot as a result of a limited time that we had to complete our robot, meaning that we did more qualitative testing and focused more on iterating our design than measuring exact performance metrics.

3. Results & Discussion

Throughout the testing of our various prototypes, there were a few consistent issues. These being power, size, grip, and maneuverability. Our final design was able to improve in all of these areas while under the control of a manual pneumatic system. The day of the final competition, the five volt pump previously used was upgraded to a 110 volt pump. This caused a significant jump in the flow rate experienced internally in the origami. Our newly designed supports were able to handle this increase in load with no fractures. Where issues did arise was the connection points between the end caps and the origami. The increased flow gave us the heightened compression needed to perform well in the course, however, This compression pulled the silicon glue in a way that left tears and small holes. Due to time constraints, some of these holes were patched with superglue. Improvements would be to design the end cap connections to work with the flow rate used to actuate, and to include air sinks to help balance the flow rate between actuators.

The strategy set in place for the competition was to attempt to complete every straight section of the course as well as the downward slope. The reason for this was the evidence in testing proving the forward motion possible, if very slow. It was also believed that the advantage of gravity would allow the robot to wiggle its way down towards the sand pit. If these could be successfully completed, partial completion of the turns and flexible tube could be attempted. We had no recorded evidence of the robot successfully completing a 90 degree turn. However, points would still be earned by getting partways through. The flexible tube was reported as containing no internal ridges. This was concerning as the design relies on the ridges for forward momentum. The exact shape, size, and rigidity of the tube was left a mystery which made it difficult to design for. Similar to the 90 degree turns, entering this section would earn partial points. If we could rely on the back feet to push the robot forward alone using the ridges before the flexible tube, we were confident that the robot could clear at least a third of this section. Finally, we did not anticipate completing the rubble, sand pit, or upwards slope. These would be the sections that would be potentially skipped if time was running short. This is due to the lack of information. The size, shape, and material of the rubble was never clear and could not be tested for. The rules regarding the sand pit changed partially through the design phase and left us little time to make significant changes to the overall design. Finally, the upwards slope would have our robot acting against gravity. In past testing, the robot tended to slide back down the tube when attempting to reach for the next ridge. With our strategy set and priorities in mind, we were ready for the final competition.

The official competition event took place on April 30th 2025. The course used can be seen in figure 2. With our first look at the rubble, flexible tube, and sand pit, we were able practice our strategy and prepare for our run. The robot struggled with the first straight section of the tubing. Alternating different actuations rapidly achieved forward motion. Part way through this section, a leak formed in one of the origami near the back end cap because of the increased flow rate of the pump. This left our third origami nearly completely nonfunctional. The leak was able to take on the brunt of the additional air flow from the new powerful pump. This left our remaining origami with a closer to ideal air flow rate to function at maximum potential in both compression and expansion. This increased functionality made up for the third origami with tactical timing and combination of activation.

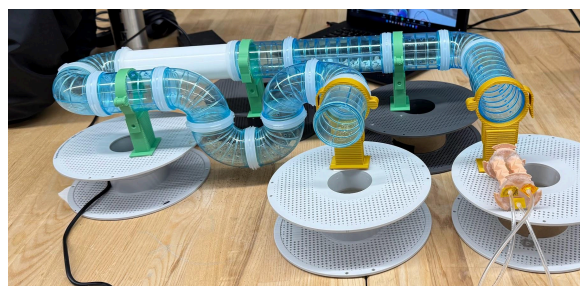


Fig. 8: An image of the competition field and our robot before the competition.

Upon reaching our medium priority of the 90 degree turn, the robot was able to find purchase in the smaller ridges. The broken actuator was positioned to the side with a wider curve, meaning it was never required to actuate to induce turning. The robot managed to make its way completely through the curve which was an unexpected result. The robot performed similarly with all other turns, including the vertical pipes. We cannot generalize our results yet as our robot contained a broken actuator in the competition testing, although it bodes well for future design iterations. Considering its performance with two working actuators, a third actuator would improve its capabilities. A future design would need to solve the leakage issue to allow for easier turning

motions in both directions. The simplest solution to this would be finding a pump of a more tolerable voltage and air flow for our binding agents.

The areas of questionable competition due to lack of knowledge were the rubble, flexible tube, and sand pit. Due to limited time, the rubble and sand pit were skipped. While it's possible that the robot may have displaced the obstacles enough to move forward, it would have taken a significant amount of time. The final shape of the robot was much larger than originally intended to make space for the triplets of origami tubing. The wide stance and adherence to the walls via the flexible feet proved useful while navigating the flexible tube. The tube was made of a material that had more friction than the rest of the course. The tube also had a slightly smaller diameter. Upon entry, the robot had no way to back up due to the high increase of contact points and friction. This made for a quick navigation of the obstacle. In future design iterations, it would be valuable to reduce the material quantity of the structure, by decreasing the width of the feet, and adding holes to the end caps in between the origami connections.

Due to the limited timeline for development, a part shortage led to complicated electrical wiring and a pump that produced an air flow too strong for the robot. If this project was to be continually iterated on, the replacement of all two way valves for three way valves would nearly halve the amount of solenoids and relays needed. This would lead to a more robust, and compact design.

Experimentation would need to be done to find the best pump for this robot. Too much power pulls on the origami and flexible silicon glue in such a way that the structure collapses onto itself at the edges and forms tears. Possible solutions to reduce this risk other than a weaker pump would be to place the edges of the origami in a secluded cavity so that they could not be pulled out from the wall of the end caps. Additionally, steps could be taken in the CAD modeling to introduce small “lips” or edges in the origami so that it connects more securely.

The robot could be developed further for real world applications. Additional sections of triplet origami that can all be actuated separately would increase the options of directionality the robot could produce. The robot may also be able to travel forward without the assistance of walls to push it along. Improvements such as these would allow this design to help explore wreckage sites, underground piping and sewers, or even become miniaturized for medical applications.

We had several findings in our design process, listed here. Firstly, pneumatic systems are very sensitive to pump flow rate, and controlling flow rates is crucial to maintaining the integrity of the structure. Secondly, having excess actuation methods in the design increases flexibility and allows the robot to function even when damaged, which is a very important thing to consider when designing robots for war, medical, and safety applications. Thirdly, CAD designs should anticipate stresses higher than planned for and should that should be accounted for in the design through additional mechanical reinforcement. Lastly, using frictional forces as a method of movement can provide ways to perform locomotion both forwards and backwards, given a specific set of actuation steps.

4. Conclusion

The final design of our robot with a manually controlled triple kresling origami pneumatic system was a success. It was able to pass all of the obstacles we listed as higher priority and exceeded our expectations for speed and consistency.

The claim we made at the start of this project to be able to create a robot that could successfully traverse all straight and curved segments was proven true. Consistent results were seen throughout the entire course with a speed that allowed us to complete our strategy to its greatest potential. The redesign of the robot to a triple origami design was paramount to this and even completing the flexible tube section. However, this redesign was done late in the design process and as such, affected our final results. It would have been nearly impossible for the robot to complete the stretch goals we had due to the diameter of the robot. In our design process, the robot required grooves at specific distances apart from one another to move forward in low friction piping. Since the flexible wall segment lacked grooves and the robot pushed through it, we could conclude that given enough frictional force, our robot might perform better without grooves, but that would require additional testing.

A limitation of our project is in its pneumatic design. Placing a pump onboard the robot and powering it would be a difficult task, leading to needed design

changes if future applications might require untethered, or mobile, soft, pipe-crawling robots.

Suggestions for future work are to improve on this design. One improvement could be to continue testing shapes for the feet, and finding which gives the best friction to the robot. Our team did not perform any frictional analyses to optimize shape or amount of surface contact.

Additional suggestions apply this design to other fields. We believe that applications in biomedical robotics exist in areas such as surgical procedures or targeted drug delivery. If the robot's size can be significantly reduced, the flow rate requirements would decrease proportionally, allowing smaller, quieter pumps that could be integrated onto the robot. The silicon-based sealants would need to be replaced with a biocompatible sealant method.

This design was an important milestone in our personal growth as engineers. We took on this challenge with the goal of learning how to design and build both a soft robotic design and a pneumatic system for the first time. The experience gained through this project and competition will be brought with us into future projects and careers. Having our minds opened to the potential of robots that can transform themselves will only make us stronger as engineers.

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